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WATER HEATER CONNECTION SYSTEM

Abstract

A water heater connection system may include: water heaters connected in parallel between a water supply pipe and a discharge pipe; and a connection controller which controls the heaters. Each heater may include: a burner which heats water flowing from the supply pipe to the heater as required; a valve which regulates a flow rate of the water flowing from the supply pipe to the heater; a sensor which detects the water flow rate flowing from the supply pipe to the heater; and an individual controller which communicates with the connection controller. The individual controller may operate in a disconnected mode when the individual controller cannot communicate with the connection controller. When the sensor does not detect water flow in the disconnected mode, the individual controller may stand by with the valve set to a first opening degree being smaller than fully open and larger than fully closed.

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Background/Summary

REFERENCE TO RELATED APPLICATIONS

[0001] This application claims priority to Japanese Patent Application No. 2024-026366 filed on Feb. 26, 2024. The entire content of the priority application is incorporated herein by reference.

TECHNICAL FIELD

[0002] The present teachings relate to a water heater connection system.

BACKGROUND ART

[0003] A water heater connection system is described in Japanese Patent Application Publication No. H06-288624. This water heater connection system includes: a plurality of water heaters disposed in parallel to each other between a water supply pipe and a hot water discharge pipe; and a connection controller configured to control operation of the plurality of water heaters. Each of the plurality of water heaters includes: a burner configured to heat water flowing from the water supply pipe to the water heater as required; a flow rate regulation valve configured to regulate a flow rate of water flowing from the water supply pipe to the water heater; a flow rate sensor configured to detect the flow rate of water flowing from the water supply pipe to the water heater; and an individual controller configured to communicate with the connection controller. The individual controller is configured to operate in a disconnected operation mode when the individual controller cannot communicate with the connection controller. When the flow rate sensor does not detect water flow in the disconnected operation mode, the individual controller stands by with the flow rate regulation valve set to fully closed.

SUMMARY

[0004] In the above-mentioned water heater connection system, since the water stops flowing from the water supply pipe into the water heater(s) operating in the disconnected operation mode, these water heater(s) cannot further be used to supply hot water. Due to this, when a part of the water heaters cannot communicate with the connection controller, performance in supplying hot water as a whole may be deteriorated in the water heater connection system. The present teachings provide an art configured to suppress deterioration in performance of a water heater connection system in supplying hot water as a whole even when some of water heaters cannot communicate with a connection controller in the water heater connection system.

[0005] In a first aspect, a water heater connection system may comprise: a plurality of water heaters connected in parallel to each other and between a water supply pipe and a hot water discharge pipe; and a connection controller configured to control operation of the plurality of water heaters. wherein each of the plurality of water heaters may include: a burner configured to heat water flowing from the water supply pipe to the water heater as required; a flow rate regulation valve configured to regulate a flow rate of the water flowing from the water supply pipe to the water heater; a flow rate sensor configured to detect the flow rate of the water flowing from the water supply pipe to the water heater; and an individual controller configured to communicate with the connection controller. the individual controller may be configured to operate in a disconnected operation mode when the individual controller cannot communicate with the connection controller. When the flow rate sensor does not detect water flow in the disconnected operation mode, the individual controller may stand by with the flow rate regulation valve set to a first opening degree which is smaller than being fully open and larger than being fully closed.

[0006] Because in the above-described water heater connection system the water heater(s) operating in the disconnected operation mode each stands by with the opening degree of the flow rate regulation valve set to the first opening degree, which is larger than being fully closed, water

continues to flow from the water supply pipe into that water heater even after the water heater has entered into the disconnected operation mode. As a result, the water heater can still be used for supplying hot water. Due to this, the performance in supplying hot water of the water heater connection system as a whole can be suppressed from deteriorating when some of the water heaters cannot communicate with the connection controller.

[0007] In a general water heater connection system, a water heater configured to operate in accordance with an instruction from a connection controller stands by with a flow rate regulation valve being fully open or fully closed. Specifically, in the water heater connection system, a water heater which the connection controller intends to operate next among water heaters that have not started operating (e.g., if none of the plurality of water heaters are operating, a water heater that is intended to be operated first) stands by with the flow rate regulation valve being fully open, and the other water heater(s) among the water heaters that have not started operating stand by with the flow rate regulation valve being fully closed. This aims to flow water concentratedly through the water heater which is intended to be operated next (e.g., the water heater that is intended to be operated first) such that that water heater can start supplying hot water immediately when the entire flow rate of the water heater connection system increases. If the water heater operating in the disconnected operation mode stands by with the opening degree of the flow rate regulation valve set to fully open, there would be less increase in the flow rate of water flowing in the water heater that is intended to be operated next (e.g., water heater that is intended to be operated first) when the entire flow rate of the water heater connection system **100** increases, as a result of which the water heater cannot start supplying hot water immediately. According to the above configuration, since the water heater operating in the disconnected operation mode stands by with the opening degree of the flow rate regulation valve set to the first opening degree that is smaller than being fully open, the flow rate of the water flowing in the water heater that is intended to be operated next (e.g., water heater that is intended to be operated first) can be significantly increased when the entire flow rate in the water heater connection system increases, as a result of which that water heater can start supplying hot water immediately.

[0008] In a second aspect, according to the first aspect, the individual controller may be configured to ignite the burner when the flow rate detected by the flow rate sensor becomes equal to or greater than a predetermined ignition flow rate. The first opening degree may be smaller than an opening degree corresponding to the ignition flow rate.

[0009] According to the above configuration, since the water heater operating in the disconnected operation mode stands by with the opening degree of the flow rate regulation valve set to an opening degree sufficiently small, the flow rate of water flowing in the water heater which the connection controller intends to operate next (e.g., water heater that is intended to be operated first) can be significantly increased when the entire flow rate in the water heater connection system increases, by which that water heater can immediately start supplying hot water.

[0010] In a third aspect, according to the second aspect, when the flow rate sensor detects water flow while the individual controller stands by with the flow rate regulation valve set to the first opening degree, the individual controller may increase an opening degree of the flow rate regulation valve from the first opening degree to a second opening degree which is larger than the opening degree corresponding to the ignition flow rate.

[0011] According to the above configuration, when the entire flow rate in the water heater connection system increases, the flow rate of the water heater operating in the disconnected operation mode can be increased, by which that water heater operating in the disconnected operation mode can immediately start supplying hot water.

[0012] In a fourth aspect, according to the third aspect, when the individual controller increases the opening degree of the flow rate regulation value to the second opening degree and the flow rate detected by the flow rate sensor increases accordingly, the individual controller may further increase the opening degree of the flow rate regulation valve. When the individual controller

increases the opening degree of the flow rate regulation valve to the second opening degree and the flow rate detected by the flow rate sensor does not increase accordingly, the individual controller may maintain the opening degree of the flow rate regulation valve.

[0013] When the flow rate that can be supplied by the water heater connection system is smaller than the flow rate that can be supplied at the hot water supply outlet(s) (i.e., when the water heater connection system is bottlenecked), by increasing the opening degree of the flow rate regulation valve, the flow rate detected by the flow rate sensor increases. Contrary to this, when the flow rate that can be supplied by the water heater connection system is greater than the flow rate that can be supplied at the hot water supply outlet(s) (i.e., when the water heater connection system is not bottlenecked), even when the opening degree of the flow rate regulation valve is increased, the flow rate detected by the flow rate sensor does not increase. According to the above configuration, the opening degree of the flow rate regulation valve can be increased to an opening degree at which the water heater connection system is not bottlenecked.

[0014] In a fifth aspect, according to the fourth aspect, when the flow rate detected by the flow rate sensor increases while the individual controller maintains the opening degree of the flow rate regulation value at the second opening degree, the individual controller may increase the opening degree of the flow rate regulation valve.

[0015] The situation where the flow rate detected by the flow rate sensor increases despite the opening degree of the flow rate regulation valve is maintained happens when the flow rate that can be supplied by the water heater connection system is greater than the flow rate that can be supplied at the hot water supply outlet(s) (i.e., the water heater connection system is not bottlenecked) and also the flow rate that can be supplied at the hot water supply outlet(s) increases. According to the above configuration, because the opening degree of the flow rate regulation valve is increased in such cases, the opening degree of the flow rate regulation valve can be increased in accordance with the increase in the flow rate that can be supplied at the hot water supply outlet(s).

[0016] In a sixth aspect, according to the fourth or fifth aspect, when the flow rate detected by the flow rate sensor decreases while the individual controller maintains the opening degree of the flow rate regulation valve at the second opening degree, the individual controller may decrease the opening degree of the flow rate regulation valve.

[0017] The situation where the flow rate detected by the flow rate sensor decreases despite the opening degree of the flow rate regulation valve is maintained happens when the flow rate that can be supplied by the water heater connection system is greater than the flow rate that can be supplied at the hot water supply outlet(s) (i.e., the water heater connection system is not bottlenecked) and also the flow rate that can be supplied at the hot water supply outlet(s) decreases. According to the above configuration, because the opening degree of the flow rate regulation valve is decreased in such cases, the opening degree of the flow rate regulation valve can be decreased in accordance with the decrease in the flow rate that can be supplied at the hot water supply outlet(s).

[0018] In a seventh aspect, according to any one of the first to sixth aspects, each of the plurality of water heaters may further comprise a notifier. When the individual controller cannot communicate with the connection controller, the individual controller may notify via the notifier that the individual controller is operating in the disconnected operation mode.

[0019] According the above configuration, when it is impossible to communicate with the connection controller and the water heater(s) are operating in the disconnected operation mode, the user can be immediately notified of that situation.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0020] FIG. 1 illustrates a configuration of a water heater connection system 100 according to an

embodiment.

[0021] FIG. **2** illustrates a water heater **1** according to the embodiment.

[0022] FIG. **3** illustrates a flowchart of a mode determination process according to the embodiment.

[0023] FIG. **4** illustrates a flowchart of a disconnected operation process of the water heater **1** according to the embodiment.

DETAILED DESCRIPTION

[0024] Representative, non-limiting examples of the present disclosure will now be described in further detail with reference to the attached drawings. This detailed description is merely intended to teach a person of skill in the art further details for practicing preferred aspects of the present teachings and is not intended to limit the scope of the disclosure. Furthermore, each of the additional features and teachings disclosed below may be utilized separately or in conjunction with other features and teachings to provide improved water heater connection systems, as well as methods for using and manufacturing the same.

[0025] Moreover, combinations of features and steps disclosed in the following detailed description may not be necessary to practice the disclosure in the broadest sense, and are instead taught merely to particularly describe representative examples of the disclosure. Furthermore, various features of the above-described and below-described representative examples, as well as the various independent and dependent claims, may be combined in ways that are not specifically and explicitly enumerated in order to provide additional useful embodiments of the present teachings.

[0026] All features disclosed in the description and/or the claims are intended to be disclosed separately and independently from each other for the purpose of original written disclosure, as well as for the purpose of restricting the claimed subject matter, independent of the compositions of the features in the embodiments and/or the claims. In addition, all value ranges or indications of groups of entities are intended to disclose every possible intermediate value or intermediate entity for the purpose of original written disclosure, as well as for the purpose of restricting the claimed subject matter.

Embodiments

[0027] As illustrated in FIG. **1**, a water heater connection system **100** according to the present embodiment comprises a plurality of water heaters **1**. The plurality of water heaters **1** is disposed in parallel to each other between a water supply pipe **110** and a hot water discharge pipe **120** such that water is supplied from the water supply pipe **110** and hot water is delivered to the hot water discharge pipe **120**. Each of the water heaters **1** is coupled via a corresponding water supply coupling pipe **112** to the water supply pipe **110**. Water such as tap water from an unillustrated water source is supplied to the water supply pipe **110**. Each of the water heaters **1** is coupled via a corresponding hot water discharge coupling pipe **122** to the hot water discharge pipe **120**. The hot water discharge pipe **120** supplies hot water to a hot water supply outlet such as a bathroom and/or a kitchen. The hot water discharge pipe **120** comprises a faucet **124** such as a tap for increasing/decreasing flow rate of hot water supplied to the supply outlet and/or shower (in FIG. **1**, described as a single faucet **124** for easier illustration). Further, each of the plurality of water heaters **1** is coupled via a corresponding gas supply coupling pipe **132** to a gas supply pipe **130**. Fuel gas such as city gas is supplied from an unillustrated gas supply source to the gas supply pipe **130**. Due to this, the plurality of water heaters **1** is supplied with fuel gas from the gas supply pipe **130**. The plurality of water heaters **1** is configured to communicate with each other via a communication cable **140**.

(Configuration of Water Heaters **1**)

[0028] As illustrated in FIG. **2**, each of the water heaters **1** comprises a burner **3**, a fan **4**, a vapor heat exchanger **6**, an inlet pipe **7**, an outlet pipe **8**, a can body **10**, a controller **30**, an input device **32**, and an indicator **34**. The burner **3** and the vapor heat exchanger **6** are accommodated within the can body **10**. The vapor heat exchanger **6** is disposed above the burner **3**. The fan **4** supplies air for

combustion from below the can body **10**. Combustion gas of the burner **3** flows within the can body **10** from bottom to top, thereby heating the vapor heat exchanger **6**. Combustion exhaust gas after heating the vapor heat exchanger **6** is discharged from an exhaust vent **10a** arranged at a top of the can body **10**.

[0029] The vapor heat exchanger **6** is coupled to the inlet pipe **7** and the outlet pipe **8**. An end of the inlet pipe **7** on an upstream side is coupled to the water supply coupling pipe **112**. An end of the outlet pipe **8** on a downstream side is coupled to the hot water discharge coupling pipe **122**. The water heater **1** further comprises a bypass pipe **9** which bypasses the vapor heat exchanger **6** and connects the inlet pipe **7** and the outlet pipe **8**, and a bypass servo **24** configured to regulate an opening degree of the bypass pipe **9** to change a bypass ratio (ratio of the flow rate of water supplied toward the bypass pipe **9** to the flow rate of water supplied to the vapor heat exchanger **6**). Water supplied from the water supply coupling pipe **112** is supplied from the inlet pipe **7** to the vapor heat exchanger **6**, and is heated by the vapor heat exchanger **6** and becomes hot water of a high temperature. Thereafter, the hot water of high temperature flowing from the vapor heat exchanger **6** is mixed with water of a low temperature flowing from the inlet pipe **7** to the bypass pipe **9**, and the temperature of the water is adjusted, and then is delivered from the outlet pipe **8** to the hot water discharge coupling pipe **122**.

[0030] The burner **3** comprises a first burner **3a**, a second burner **3b**, and a third burner **3c**. The burner **3** is supplied by a gas pipe **11** with the fuel gas. An upstream end of the gas pipe **11** is coupled to the gas supply coupling pipe **132**. The gas pipe **11** comprises a main gas electromagnetic valve **12**, a gas proportional valve **13**, and switch gas electromagnetic valves **14a**, **14b**, **14c** in this order from the upstream side. When the main gas electromagnetic valve **12** is in the open state, the fuel gas is supplied to the burner **3**, whereas when the main gas electromagnetic valve **12** is in the closed state, the fuel gas supply to the burner **3** is stopped. The gas proportional valve **13** is configured to regulate a volume of the fuel gas supplied to the burner **3**. The switch gas electromagnetic valves **14a**, **14b**, **14c** are disposed corresponding to the first burner **3a**, the second burner **3b**, and the third burner **3c**, respectively. When each of the switch gas electromagnetic valves **14a**, **14b**, **14c** is in the open state, the fuel gas is supplied to the corresponding one of the first burner **3a**, the second burner **3b**, and the third burner **3c**, whereas when each of the switch gas electromagnetic valves **14a**, **14b**, **14c** is in the closed state, the fuel gas stops being supplied to the corresponding one of the first burner **3a**, the second burner **3b**, and the third burner **3c**. By switching each of the switch gas electromagnetic valves **14a**, **14b**, **14c** between the open state and the closed state, a range of combustion (combustion range) of the burner **3** can be switched.

[0031] In each water heater **1**, the combustion range of the burner **3** can be switched among three levels. A first combustion range being at the lowest level corresponds to a state in which the first burner **3a** operates and the second and third burners **3b**, **3c** are not operated. A second combustion range being at a higher level than the first combustion range corresponds to a state in which the first and second burners **3a**, **3b** operate and the third burner **3c** is not operated. A third combustion range being at a higher level than the second combustion range and being at the highest level corresponds to a state in which all the first burner **3a**, the second burner **3b**, and the third burner **3c** operate. In the water heater **1**, a rotation speed of the fan **4** is predetermined according to a current combustion range of the burner **3** and the combustion level within that combustion range, and thus the fan **4** rotates at the predetermined rotation speed.

[0032] In proximity to the burner **3**, an ignition plug **16** for igniting the burner **3** and a flame rod **17** configured to detect combustion flame of the burner **3** are disposed. Further, an igniter **15** configured to apply high voltage on the ignition plug **16** to generate spark discharge is disposed.

[0033] The inlet pipe **7** comprises a flow rate sensor **18** configured to detect a flow rate of water supplied to the inlet pipe **7** (which is equal to a flow rate of hot water discharged from the outlet pipe **8**), a flow rate regulation valve **19** configured to regulate the flow rate of the water supplied to the inlet pipe **7**, and a supplied water temperature sensor **25** configured to detect a temperature of

the water supplied to the inlet pipe **7**. The outlet pipe **8** comprises a heat exchanger outlet temperature sensor **21** configured to detect the temperature of the water flowing from the vapor heat exchanger **6** and a discharged hot water temperature sensor **23** configured to detect the temperature of the water flowing from the outlet pipe **8**.

[0034] The controller **30** is configured to control operation of the water heater **1**. The controller **30** is an electrical unit composed of a microcomputer, a memory, etc. The controller **30** operates in accordance with a program stored in the memory. The controller **30** is inputted with detected signals of the flame rod **17**, the flow rate sensor **18**, the supplied water temperature sensor **25**, the heat exchanger outlet temperature sensor **21**, and the discharged hot water temperature sensor **23**. Further, operations of the fan **4**, the main gas electromagnetic valve **12**, the gas proportional valve **13**, the switch gas electromagnetic valves **14a**, **14b**, **14c**, the igniter **15**, the flow rate regulation valve **19**, and the bypass servo **24** are controlled by control signals outputted from the controller **30**. The controller **30** is coupled with the input device **32** and the indicator **34**. The input device **32** comprises for example a plurality of input switches. A user of the water heater **1** is able to input various settings for the operation of the water heater **1** via the input device **32**. The indicator **34** comprises for example a liquid crystal display. The indicator **34** is configured to notify the user of the water heater **1** of various information related to the operation of the water heater **1**. The controller **30** is further coupled with the communication cable **140**. The controller **30** is configured to communicate bidirectionally with the controllers **30** of the other water heaters **1** via the communication cable **140**.

(Ignition Process and Extinguish Process)

[0035] The controller **30** constantly monitors the flow rate detected by the flow rate sensor **18**. When the flow rate detected by the flow rate sensor **18** becomes equal to or more than a predetermined ignition flow rate (e.g., 2.7 L/min) with the burner **3** not operating, the controller **30** executes the ignition process of the burner **3**. That is, the controller **30** activates the fan **4**, sets the main gas electromagnetic valve **12** and the switch gas electromagnetic valve **14a** in the open state, activates the igniter **15**, and ignites the first burner **3a** to start the combustion of the burner **3**.

[0036] While the burner **3** is operating, the controller **30** regulates opening/closing of the switch gas electromagnetic valves **14a**, **14b**, **14c**, an opening degree of the gas proportional valve **13**, the rotation speed of the fan **4**, and an opening degree of the bypass servo **24** such that the temperature detected by the heat exchanger outlet temperature sensor **21** becomes a predetermined target temperature (e.g., 60° C.) and the temperature detected by the discharged hot water temperature sensor **23** becomes a supplied hot water designated temperature designated by the user.

[0037] When the flow rate detected by the flow rate sensor **18** becomes less than a predetermined extinguish flow rate (e.g., 2.0 L/min) while the burner **3** is operating, the controller **30** executes an extinguish process of the burner **3**. That is, the controller **30** stops driving the fan **4**, sets the main gas electromagnetic valve **12** and the switch gas electromagnetic valves **14a**, **14b**, **14c** in the closed state, and extinguishes the burner **3** to finish the combustion of the burner **3**.

(Solo Operation Mode and Connected Operation Mode)

[0038] The water heater **1** is configured to select either a solo operation mode or a connected operation mode as its operation mode. In the solo operation mode, the water heater **1** operates without communicating with other water heater(s) **1**. In the connected operation mode, the water heater **1** operates by utilizing communication with the other water heaters **1**. When the connected operation mode is selected as the operation mode, the water heater **1** is configured to select either a parent machine operation mode or a child machine operation mode as its operation mode. In the parent machine operation mode, the water heater **1** operates dominantly relative to the other water heaters **1**. In the child machine operation mode, the water heater **1** operates subjectively relative to the other water heater(s) **1**. Specifically, the water heater **1** selected to be in the parent machine operation mode (may be referred to as simply water heater **1** being a parent machine) sends an operation instruction to another water heater **1** selected to operate in the child machine operation

mode (may be referred to as simply water heater **1** being a child machine). The water heater **1** being the child machine (hereafter, child water heater **1**) operates in accordance with the operation instruction received from the water heater **1** being the parent machine (hereafter, parent water heater **1**). That is, when the water heater **1** is selected to operate in the parent machine operation mode, the controller **30** of that water heater **1** functions as a connection controller configured to control respective operations of the plural water heaters **1** and also as an individual controller configured to control an operation of the water heater **1** itself. Further if the water heater **1** is selected to operate in the child machine operation mode, the controller **30** of that water heater **1** does not function as the connection controller configured to control the respective operations of the plural water heaters **1**, but can function only as the individual controller configured to control the operation of that water heater **1** itself. Whether the operation mode of the water heater **1** is to operate in the solo operation mode or in the connected operation mode can be inputted in advance via the input device **32**. Further, if the connected operation mode is selected as the operation mode, whether the operation mode of the water heater **1** is the parent machine operation mode or the child machine operation mode can be inputted in advance via the input device **32**.

(Mode Determination Process)

[0039] When power of the water heater **1** is turned on, the controller **30** executes a mode determination process shown in FIG. **3**.

[0040] In **S2**, the controller **30** determines whether the operation mode of the water heater **1** is the connected operation mode or not. If the operation mode of the water heater **1** is the solo operation mode (in case of NO), the process proceeds to **S4**.

[0041] In **S4**, the controller **30** executes processes in the solo operation mode. In the solo operation mode, the controller **30** stands by with the flow rate regulation valve **19** set to fully open.

[0042] If a mode setting of the water heater **1** is the connected operation mode in **S2** (in case of YES), the process proceeds to **S6**. In **S6**, the controller **30** determines whether the operation mode of the water heater **1** is the parent machine operation mode or not. If the operation mode is the parent machine operation mode (in case of YES), the process proceeds to **S8**.

[0043] In **S8**, the controller **30** executes a parent machine operation process in order to cause the water heater **1** to operate in the parent machine operation mode. That is, the controller **30** utilizes the communication with other child water heater(s) **1** to determine whether the flow rate regulation valve **19** of each water heater **1** is to be fully open or fully closed, and instructs the child water heater(s) **1** to stand by with the flow rate regulation valve **19** being fully open or fully closed.

[0044] When the operation mode is the child machine operation mode in **S6** (in case of NO), the process proceeds to **S10**. In **S10**, the controller **30** determines whether it is possible to communicate with the parent water heater **1**. If it is possible to communicate with the parent water heater **1** (in case of YES), the process proceeds to **S12**. In **S12**, the controller **30** executes a child machine operation process in order to cause the water heater **1** to operate in the child machine operation mode. That is, the controller **30** utilizes the communication with the parent water heater **1** to stand by with the flow rate regulation valve **19** being fully open or fully closed in accordance with an instruction from the parent water heater **1**.

[0045] When it is impossible to communicate with the parent water heater **1** in **S10** (in case of NO), the process proceeds to **S14**. In **S14**, the controller **30** executes a disconnected operation process (FIG. **4**) in order to cause the water heater **1** to operate in the disconnected operation mode. (Disconnected Operation Process)

[0046] As illustrated in FIG. **4**, in **S20**, the controller **30** notifies the user of being operating in the disconnected operation mode via the indicator **34**.

[0047] In **S22**, the controller **30** stands by with an opening degree of the flow rate regulation valve **19** set to a first opening degree. The first opening degree is larger than 0% and smaller than 100% when the opening degree of being fully closed is deemed 0% and the opening degree of being fully open is deemed 100%. Further, the first opening degree is smaller than an opening degree

corresponding to the ignition flow rate (e.g., 2.7 L/min) (that is, opening degree of the flow rate regulation valve **19** at which the flow rate of water flowing in the flow rate regulation valve **19** becomes the ignition flow rate when the water is supplied to the flow rate regulation valve **19** with a sufficiently great supplying pressure).

[0048] In **S24**, the controller **30** stands by until water flow starts in the water heater **1**. The controller **30** determines that the water flow has started when the flow rate detected by the flow rate sensor **18** remains equal to or more than a predetermined water flow detection flow rate (e.g., 1.0 L/min) over a predetermined time period (e.g., one minute) or longer. Alternatively, the controller **30** may determine that the water flow has started when an accumulated flow rate since when the water flow started to be detected by the flow rate sensor **18** becomes equal to or more than a water flow accumulated flow rate (e.g., 1.0 L). When it is determined that the water flow has started in **S24** (in case of YES), the process proceeds to **S26**.

[0049] In **S26**, the controller **30** increases the opening degree of the flow rate regulation valve **19** by a predetermined opening degree increase amount (e.g., 5%).

[0050] In **S28**, the controller **30** determines whether the flow rate detected by the flow rate sensor **18** increases by a predetermined flow rate (e.g. 0.2 L/min) or more. When it is detected that the detected flow rate increases by the predetermined flow rate (e.g., 0.2 L/min) or more (in case of YES), the process returns to **S26**, and the controller **30** further increases the opening degree of the flow rate regulation valve **19**. When it is not detected that the detected flow rate increases by the predetermined flow rate (e.g., 0.2 L/min) or more in **S28** (in case of NO), the process proceeds to **S30**.

[0051] If, at the time the process of **S26** is performed, the flow rate that can be supplied by the water heater connection system **100** is smaller than the flow rate that can be supplied by a faucet **124** of the hot water discharge pipe **120** (i.e., when the water heater connection system **100** is bottlenecked), the flow rate detected by the flow rate sensor **18** in **S28** increases by increasing the opening degree of the flow rate regulation valve **19** in **S26**. Contrary to this, if, at the time the process of **S26** is performed, the flow rate that can be supplied by the water heater connection system **100** is greater than the flow rate that can be supplied by the faucet **124** of the hot water discharge pipe **120** (i.e., when the water heater connection system **100** is not bottlenecked), even when the opening degree of the flow rate regulation valve **19** is increased in **S26**, the flow rate detected by the flow rate sensor **18** in **S28** does not increase. Due to this, by conducting the processes of **S26** and **S28** repeatedly, the controller **30** can increase the opening degree of the flow rate regulation valve **19** to an opening degree at which the water heater connection system **100** would not be bottlenecked.

[0052] In **S30**, the controller **30** stands by until a predetermined time (e.g., one minute) elapses.

[0053] In **S32**, the controller **30** determines whether the flow rate detected by the flow rate sensor **18** increases by the predetermined flow rate (e.g., 0.2 L/min) or more. When it is detected that the detected flow rate increases by the predetermined flow rate (e.g., 0.2 L/min) or more (in case of YES), the process proceeds to **S34**. In **S34**, the controller **30** increases the opening degree of the flow rate regulation valve **19** by the predetermined opening degree increase amount (e.g., 5%). After **S34**, the process returns to **S30**. When it is not detected that the detected flow rate increases by the predetermined flow rate (e.g., 0.2 L/min) or more in **S32** (in case of NO), the process proceeds to **S36**.

[0054] Since the process of **S32** is performed after the process of **S30**, at the time the process of **S32** is executed, the opening degree of the flow rate regulation valve **19** was not increased or decreased immediately before then. Due to this, the situation where the flow rate detected by the flow rate sensor **18** increases in **S32** happens when the flow rate that can be supplied by the water heater connection system **100** is greater than the flow rate that can be supplied by the faucet **124** of the hot water discharge pipe **120** (that is, the water heater connection system **100** is not bottlenecked) and also the flow rate that can be supplied by the faucet **124** of the hot water

discharge pipe **120** increases. Due to this, by executing the processes of **S32** and **S34**, the controller **30** can increase the opening degree of the flow rate regulation valve **19** in accordance with the increase in the flow rate that can be supplied by the faucet **124** of the hot water discharge pipe **120**. [0055] In **S36**, the controller **30** determines whether the flow rate detected by the flow rate sensor **18** has decreased by the predetermined flow rate (e.g., 0.2 L/min) or more. When it is detected that the detected flow rate has decreased by the predetermined flow rate (e.g., 0.2 L/min) or more (in case of YES), the process proceeds to **S38**. In **S38**, the controller **30** decreases the opening degree of the flow rate regulation valve **19** by a predetermined opening degree decrease amount (e.g., 5%). After **S38**, the process returns to **S30**. When it is not detected that the detected flow rate has decreased by the predetermined flow rate (e.g., 0.2 L/min) or more in **S36** (in case of NO), the process proceeds to **S40**.

[0056] Since the process of **S36** is performed after the processes of **S30** and **S32**, at the time the process of **S36** is executed, the flow rate regulation valve **19** has not gone through any increase nor decrease of its opening degree immediately before **S36**. Due to this, the situation where the flow rate detected by the flow rate sensor **18** decreases in **S36** happens when the flow rate that can be supplied by the water heater connection system **100** is greater than the flow rate that can be supplied by the faucet **124** of the hot water discharge pipe **120** (that is, the water heater connection system **100** is not bottlenecked) and the flow rate that can be supplied by the faucet **124** of the hot water discharge pipe **120** has decreased. Due to this, by executing the processes of **S36** and **S38**, the controller **30** can decrease the opening degree of the flow rate regulation valve **19** in accordance with the decrease in the flow rate that can be supplied by the faucet **124** of the hot water discharge pipe **120**.

[0057] In **S40**, the controller **30** determines whether the water flow has finished in the water heater **1**. The controller **30** determines that the water flow has finished when the flow rate detected by the flow rate sensor **18** is less than the predetermined water flow detection flow rate (e.g., 1.0 L/min) over a predetermined time period (e.g., one minute) or longer. When the water flow has not finished in **S40** (in case of NO), the process proceeds to **S32**. When the water flow has finished in **S40** (in case of YES), the process proceeds to **S22**.

Modifications

[0058] In the above embodiments, a configuration where one of the plurality of water heaters **1** operates as the parent machine and the other water heaters **1** operate as the child machines was described. Differently from the above embodiment, an unillustrated controller dedicated for connection control may be coupled to the communication cable **140**, by which all the water heaters **1** of the plurality of water heaters **1** may operate in accordance with an instruction from the controller dedicated for connection control. In this case, each water heater **1** of the plurality of water heaters **1** may be configured to select only the child machine operation mode as an operation mode when the connected operation mode is selected.

Features of Embodiment

[0059] As described above, in one or more embodiments, the water heater connection system **100** comprises the plurality of water heaters **1** connected in parallel to each other and between the water supply pipe **110** and the hot water discharge pipe **120**; and the controller **30** of the parent water heater **1** (example for a connection controller) configured to control operation of the plurality of water heaters **1**. Each of the plurality of water heaters **1** includes: the burner **3** configured to heat water flowing from the water supply pipe **110** to the water heater **1** as required; the flow rate regulation valve **19** configured to regulate the flow rate of water flowing from the water supply pipe **110** to the water heater **1**; the flow rate sensor **18** configured to detect the flow rate of water flowing from the water supply pipe **110** to the water heater **1**; and the controller **30** (example for an individual controller) configured to communicate with the controller **30** of the water heater **1** being the parent machine. The controller **30** is configured to operate in the disconnected operation mode when the controller **30** cannot communicate with the controller **30** of the water heater **1** being the

parent machine. When the flow rate sensor **18** does not detect water flow in the disconnected operation mode, the controller **30** stands by with the opening degree of the flow rate regulation valve **19** set to the first opening degree which is smaller than being fully open and larger than being fully closed.

[0060] Because in the above-described water heater connection system **100** the water heater(s) **1** operating in the disconnected operation mode each stands by with the opening degree of the flow rate regulation valve **19** set to the first opening degree which is larger than being fully closed, water continues to flow from the water supply pipe **110** into that water heater **1** even after the water heater **1** has entered into the disconnected operation mode. As a result of this, the water heater **1** can still be used for supplying hot water. Due to this, the performance in supplying hot water of the water heater connection system **100** as a whole can be suppressed from deteriorating when one or more of the water heaters **1** cannot communicate with the controller **30** of the parent water heater **1**.

[0061] In the general water heater connection system **100**, the water heater **1** configured to operate in accordance with an instruction from the controller **30** of the parent water heater **1** stands by with the flow rate regulation valve **19** set to fully open or fully closed. Specifically, in the water heater connection system **100**, the water heater **1** which the controller **30** of the parent water heater **1** intends to operate next among the water heaters **1** that have not started operating (e.g., if none of the plurality of water heaters **1** are operating, the water heater **1** that is intended to be operated first) stands by with the flow rate regulation valve **19** set to fully open, and the other water heaters **1** among the water heaters **1** that have not started operating stand by with the flow rate regulation valves **19** being fully closed. This aims to flow water concentratedly through the water heater **1** which is intended to be operated next (e.g., the water heater **1** that is intended to be operated first) such that that water heater **1** can start supplying hot water immediately when the entire flow rate of the water heater connection system **100** increases. If the water heater **1** operating in the disconnected operation mode stands by with the opening degree of the flow rate regulation valve **19** set to fully open, there would be less increase in the flow rate of water flowing in the water heater **1** that is intended to be operated next (e.g., water heater **1** that is intended to be operated first) when the entire flow rate of the water heater connection system **100** increases, as a result of which the water heater **1** cannot start supplying hot water immediately. According to the above configuration, since the water heater **1** operating in the disconnected operation mode stands by with the opening degree of the flow rate regulation valve **19** set to the first opening degree that is smaller than being fully open, the flow rate of the water flowing in the water heater **1** that is intended to be operated next (e.g., water heater **1** that is intended to be operated first) can be significantly increased when the entire flow rate in the water heater connection system **100** increases, as a result of which that water heater **1** can start supplying hot water immediately.

[0062] In one or more embodiments, the controller **30** is configured to ignite the burner **3** when the flow rate detected by the flow rate sensor **18** is equal to or greater than the predetermined ignition flow rate. The first opening degree is smaller than an opening degree corresponding to the ignition flow rate.

[0063] According to the above configuration, since the water heater **1** operating in the disconnected operation mode stands by with the opening degree of the flow rate regulation valve **19** set to an opening degree sufficiently small, the flow rate of water flowing in the water heater **1** that is intended to be operated next (e.g., water heater **1** that is intended to be operated first) can be significantly increased when the entire flow rate in the water heater connection system **100** increases, by which that water heater **1** can immediately start supplying hot water.

[0064] In one or more embodiments, when the flow rate sensor **18** detects water flow while the controller **30** stands by with the flow rate regulation valve **19** set to the first opening degree, the controller **30** increases the opening degree of the flow rate regulation valve **19** from the first opening degree to the second opening degree which is larger than the opening degree corresponding to the ignition flow rate.

[0065] According to the above configuration, when the entire flow rate in the water heater connection system **100** increases, the flow rate of the water heater **1** operating in the disconnected operation mode can be immediately increased, by which that water heater **1** operating in the disconnected operation mode can immediately start supplying hot water.

[0066] In one or more embodiments, when the controller **30** increases the opening degree of the flow rate regulation value **19** to the second opening degree and the flow rate detected by the flow rate sensor **18** increases accordingly, the controller **30** further increases the opening degree of the flow rate regulation valve **19**. When the controller **30** increases the opening degree of the flow rate regulation value **19** to the second opening degree and the flow rate detected by the flow rate sensor **18** does not increase accordingly, the controller **30** maintains the opening degree of the flow rate regulation valve **19**.

[0067] When the flow rate that can be supplied by the water heater connection system **100** is smaller than the flow rate that can be supplied at the hot water supply outlet(s) (i.e., when the water heater connection system **100** is bottlenecked), by increasing the opening degree of the flow rate regulation valve **19**, the flow rate detected by the flow rate sensor **18** increases. Contrary to this, when the flow rate that can be supplied by the water heater connection system **100** is greater than the flow rate that can be supplied at the hot water supply outlet(s) (i.e., when the water heater connection system **100** is not bottlenecked), even when the opening degree of the flow rate regulation valve **19** is increased, the flow rate detected by the flow rate sensor **18** does not increase. According to the above configuration, the opening degree of the flow rate regulation valve **19** can be increased to an opening degree at which the water heater connection system **100** is not bottlenecked.

[0068] In one or more embodiments, when the flow rate detected by the flow rate sensor **18** increases while the controller **30** maintains the opening degree of the flow rate regulation value **19** at the second opening degree, the controller **30** increases the opening degree of the flow rate regulation valve **19**.

[0069] The situation where the flow rate detected by the flow rate sensor **18** increases despite the opening degree of the flow rate regulation valve **19** is maintained happens when the flow rate that can be supplied by the water heater connection system **100** is greater than the flow rate that can be supplied at the hot water supply outlet(s) (i.e., the water heater connection system **100** is not bottlenecked) and also the flow rate that can be supplied at the hot water supply outlet(s) increases. According to the above configuration, because the opening degree of the flow rate regulation valve **19** is increased in such cases, the opening degree of the flow rate regulation valve **19** can be increased in accordance with the increase in the flow rate that can be supplied at the hot water supply outlet(s).

[0070] In one or more embodiments, when the flow rate detected by the flow rate sensor **18** decreases while the controller **30** maintains the opening degree of the flow rate regulation valve **19** at the second opening degree, the controller **30** decreases the opening degree of the flow rate regulation valve **19**.

[0071] The situation where the flow rate detected by the flow rate sensor **18** decreases despite the opening degree of the flow rate regulation valve **19** is maintained happens when the flow rate that can be supplied by the water heater connection system **100** is greater than the flow rate that can be supplied at the hot water supply outlet(s) (i.e., the water heater connection system **100** is not bottlenecked) and also the flow rate that can be supplied at the hot water supply outlet(s) decreases. According to the above configuration, because the opening degree of the flow rate regulation valve **19** is decreased in such cases, the opening degree of the flow rate regulation valve **19** can be decreased in accordance with the decrease in the flow rate that can be supplied at the hot water supply outlet(s).

[0072] In one or more embodiments, each of the plurality of water heaters **1** further comprises the indicator **34** (example for a notifier). When the controller **30** cannot communicate with the

controller **30** of the parent water heater **1**, the controller **30** notifies via the indicator **34** that the controller **30** is operating in the disconnected operation mode.

[0073] According the above configuration, when it is impossible to communicate with the controller **30** of the parent water heater **1** and the water heater(s) **1** are operating in the disconnected operation mode, the user can be immediately notified of that situation.

Claims

1. A water heater connection system comprising: a plurality of water heaters connected in parallel to each other and between a water supply pipe and a hot water discharge pipe; and a connection controller configured to control operation of the plurality of water heaters, wherein each of the plurality of water heaters includes: a burner configured to heat water flowing from the water supply pipe to the water heater as required; a flow rate regulation valve configured to regulate a flow rate of the water flowing from the water supply pipe to the water heater; a flow rate sensor configured to detect the flow rate of the water flowing from the water supply pipe to the water heater; and an individual controller configured to communicate with the connection controller, the individual controller is configured to operate in a disconnected operation mode when the individual controller cannot communicate with the connection controller, and when the flow rate sensor does not detect water flow in the disconnected operation mode, the individual controller stands by with the flow rate regulation valve set to a first opening degree which is smaller than being fully open and larger than being fully closed.
 2. The water heater connection system according to claim 1, wherein the individual controller is configured to ignite the burner when the flow rate detected by the flow rate sensor is equal to or greater than a predetermined ignition flow rate, and the first opening degree is smaller than an opening degree corresponding to the ignition flow rate.
 3. The water heater connection system according to claim 2, wherein, when the flow rate sensor detects water flow while the individual controller stands by with the flow rate regulation valve set to the first opening degree, the individual controller increases the opening degree of the flow rate regulation valve from the first opening degree to a second opening degree which is larger than the opening degree corresponding to the ignition flow rate.
 4. The water heater connection system according to claim 3, wherein, when the individual controller increases the opening degree of the flow rate regulation value to the second opening degree and the flow rate detected by the flow rate sensor increases accordingly, the individual controller further increases the opening degree of the flow rate regulation valve, and when the individual controller increases the opening degree of the flow rate regulation value to the second opening degree and the flow rate detected by the flow rate sensor does not increase accordingly, the individual controller maintains the opening degree of the flow rate regulation valve.
 5. The water heater connection system according to claim 4, wherein, when the flow rate detected by the flow rate sensor increases while the individual controller maintains the opening degree of the flow rate regulation value at the second opening degree, the individual controller increases the opening degree of the flow rate regulation valve.
 6. The water heater connection system according to claim 4, wherein, when the flow rate detected by the flow rate sensor decreases while the individual controller maintains the opening degree of the flow rate regulation valve at the second opening degree, the individual controller decreases the opening degree of the flow rate regulation valve.
 7. The water heater connection system according to claim 1, wherein each of the plurality of water heaters further comprises a notifier, when the individual controller cannot communicate with the connection controller, the individual controller notifies via the notifier that the individual controller is operating in the disconnected operation mode.
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